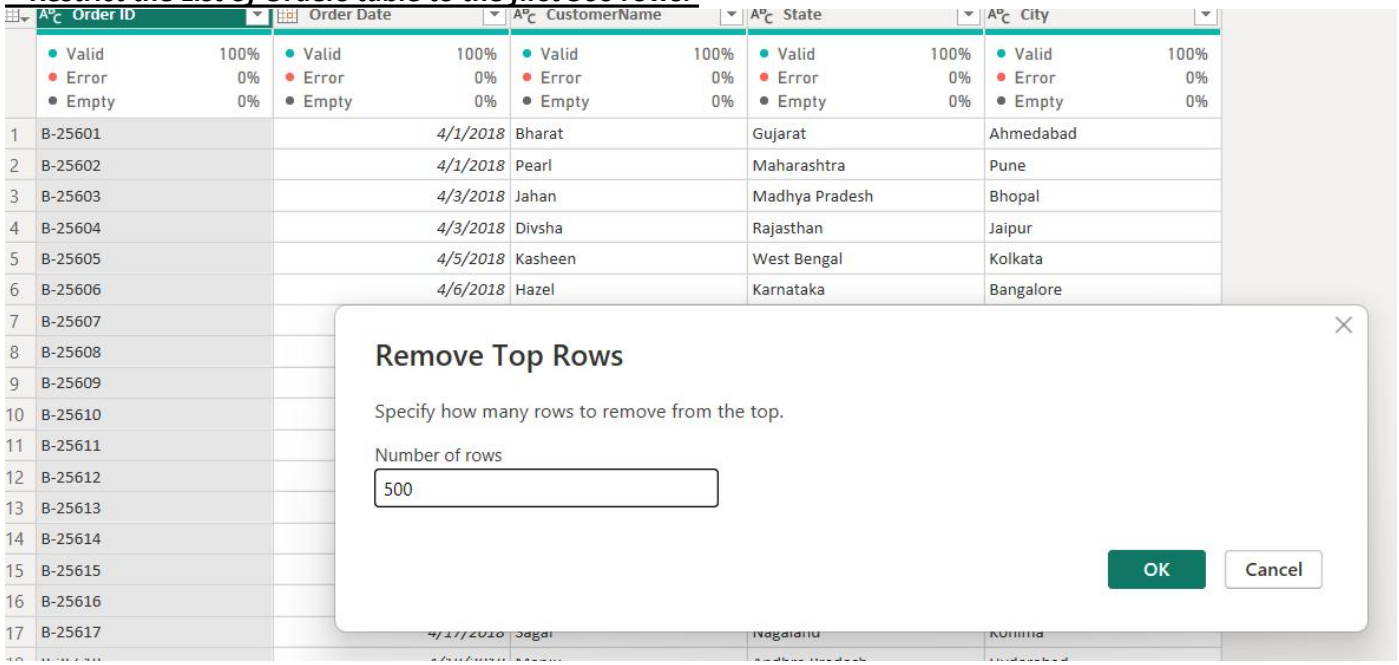
Data Analytics (DA)
Module -2
Power BI Assignment 1 – Data Transformation & Data Modeling

Tasks
Data Import & Transformation
1. Data Import:

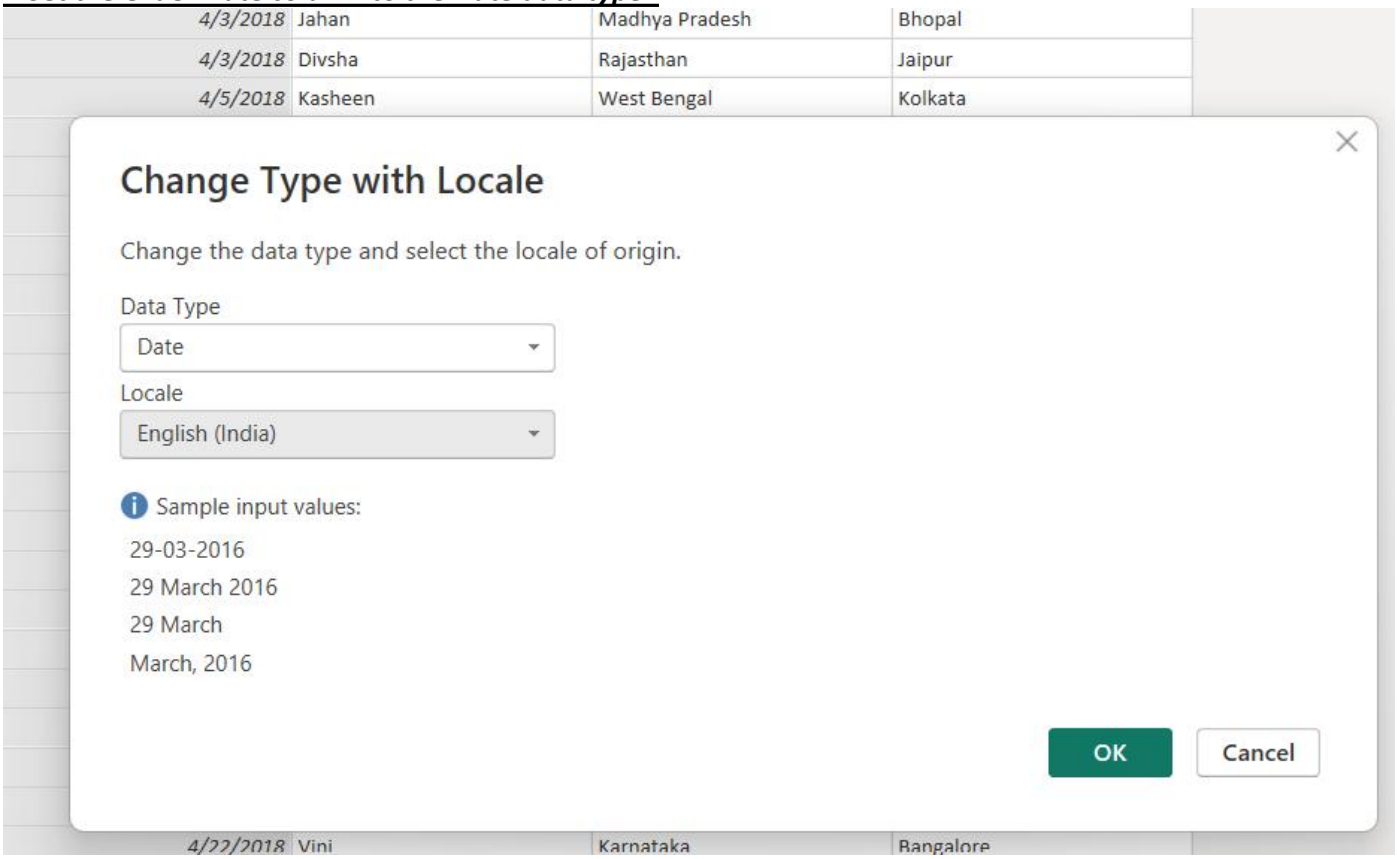
- Import *List of Orders.csv* into Power BI.
- Open *the list of orders* in **the Power Query Editor** using the **Transform Data** option.
- Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.

2. Row Limiting & Data Types:

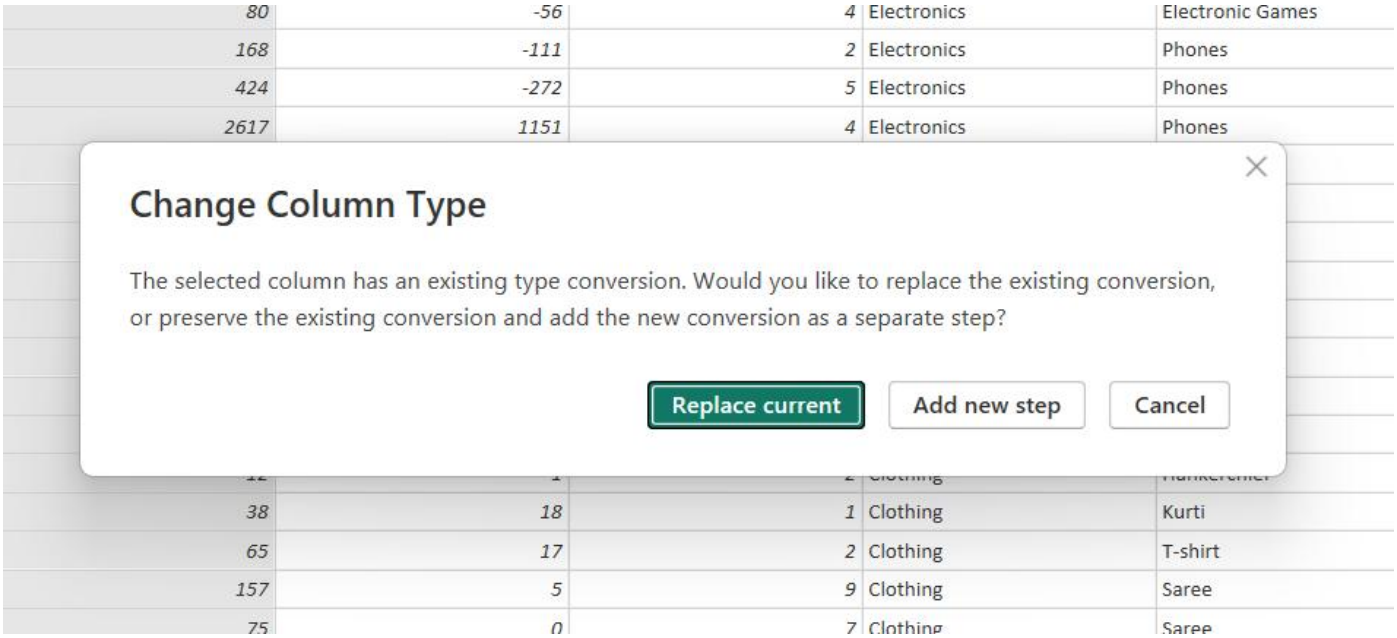
○ Restrict the List of Orders table to the first 500 rows.



○ Set the Order Date column to the Date data type.



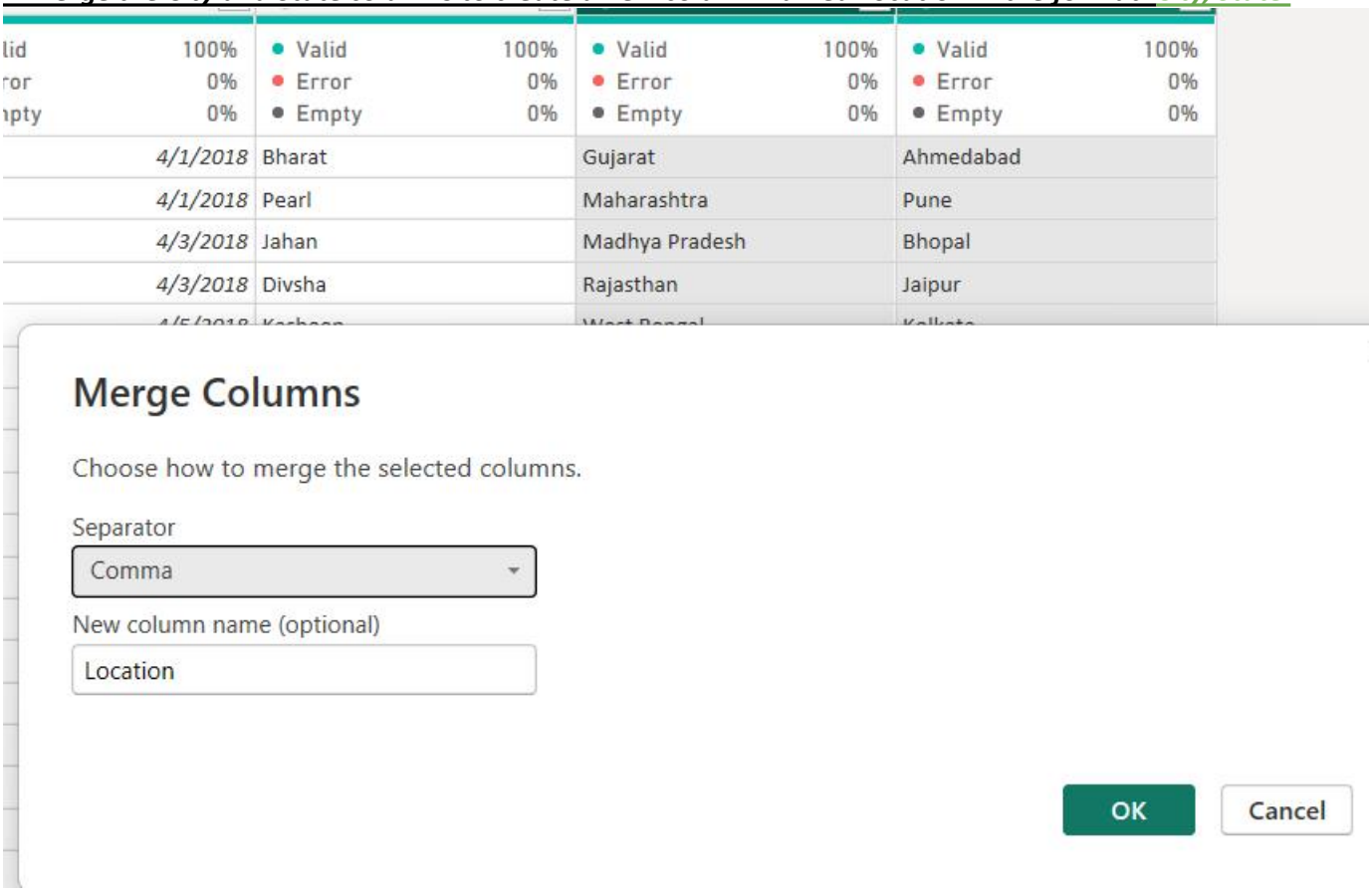
○ Change the Amount and Target columns to fixed decimal numbers.



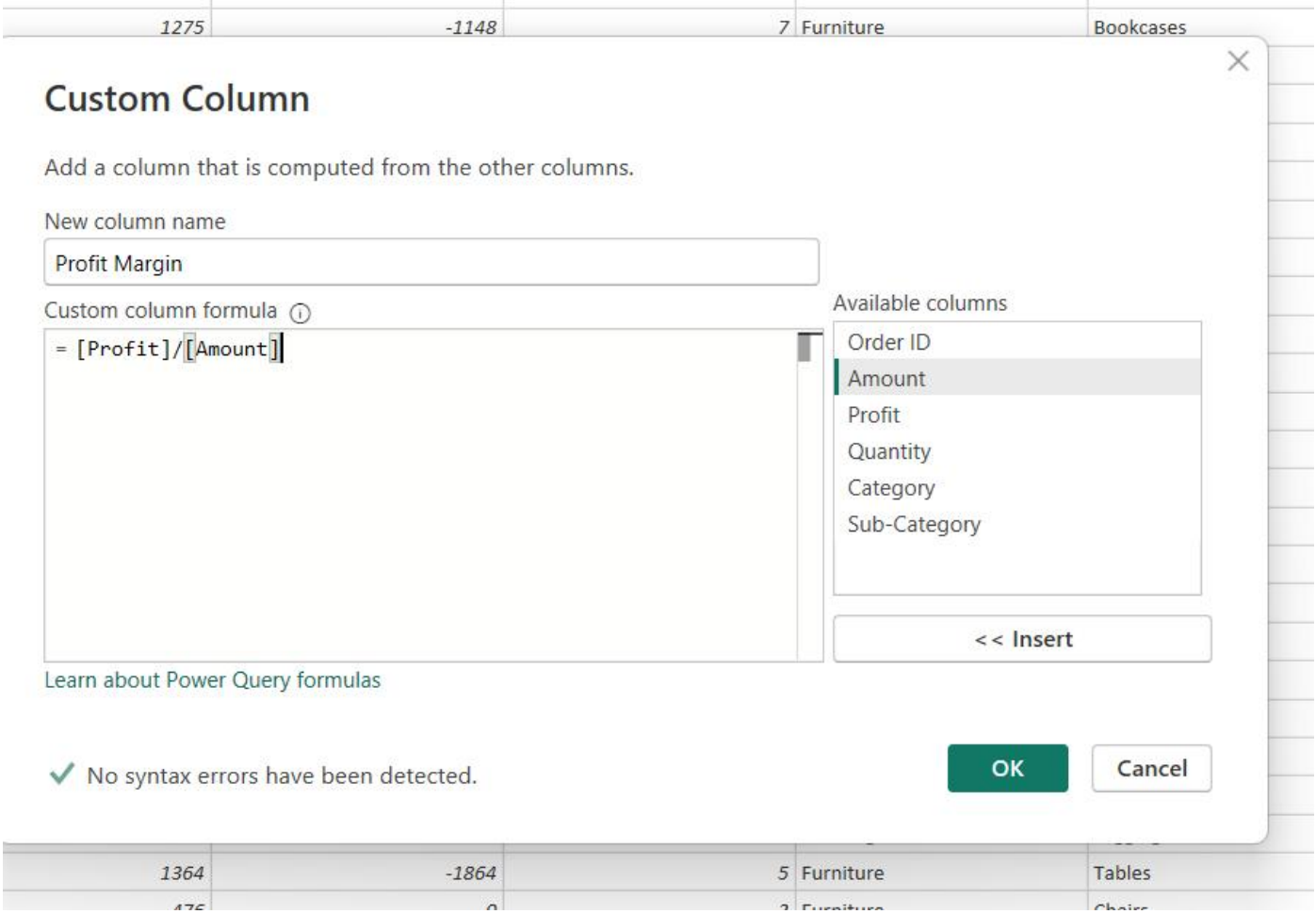
3. Text Formatting:

4. Column Creation:

○ ***Merge the City and State columns to create a new column named Location in the format: City, State***



○ ***Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)***

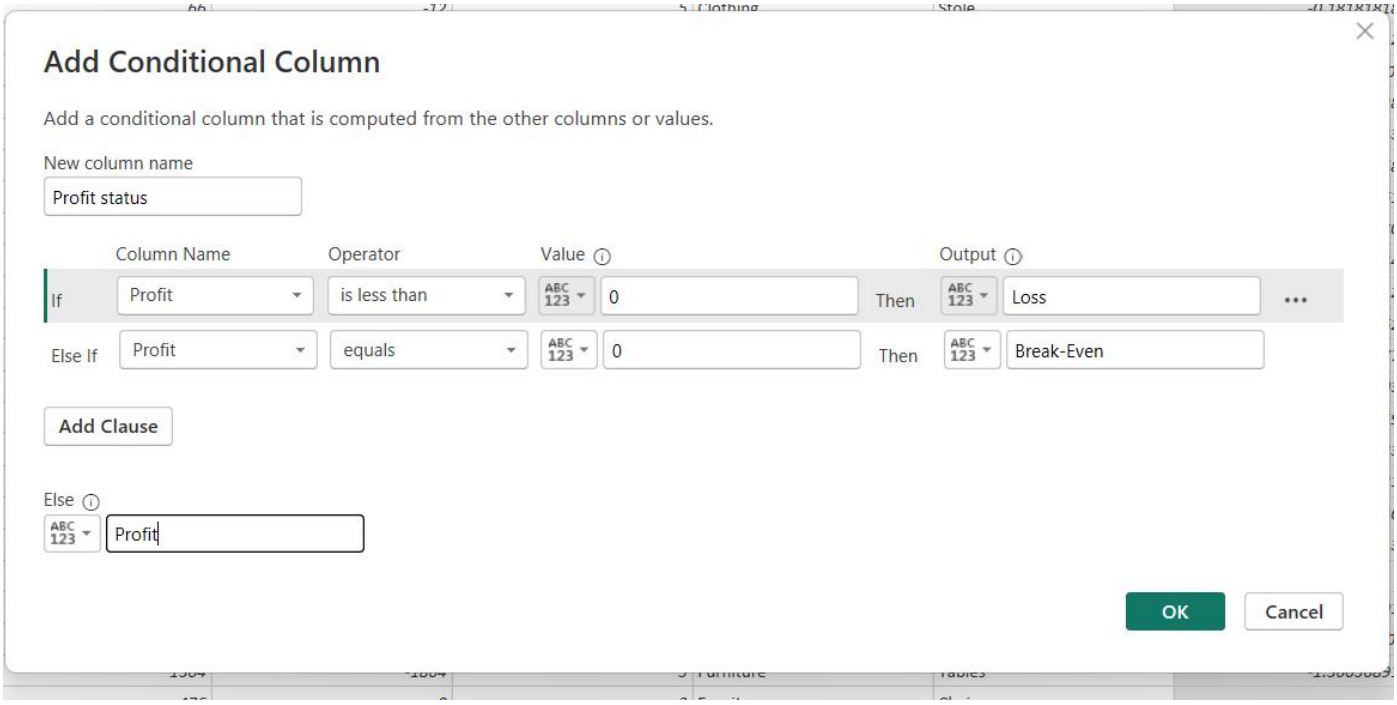


5. Conditional Column:

○ Add a conditional column named Profit Status based on: ■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

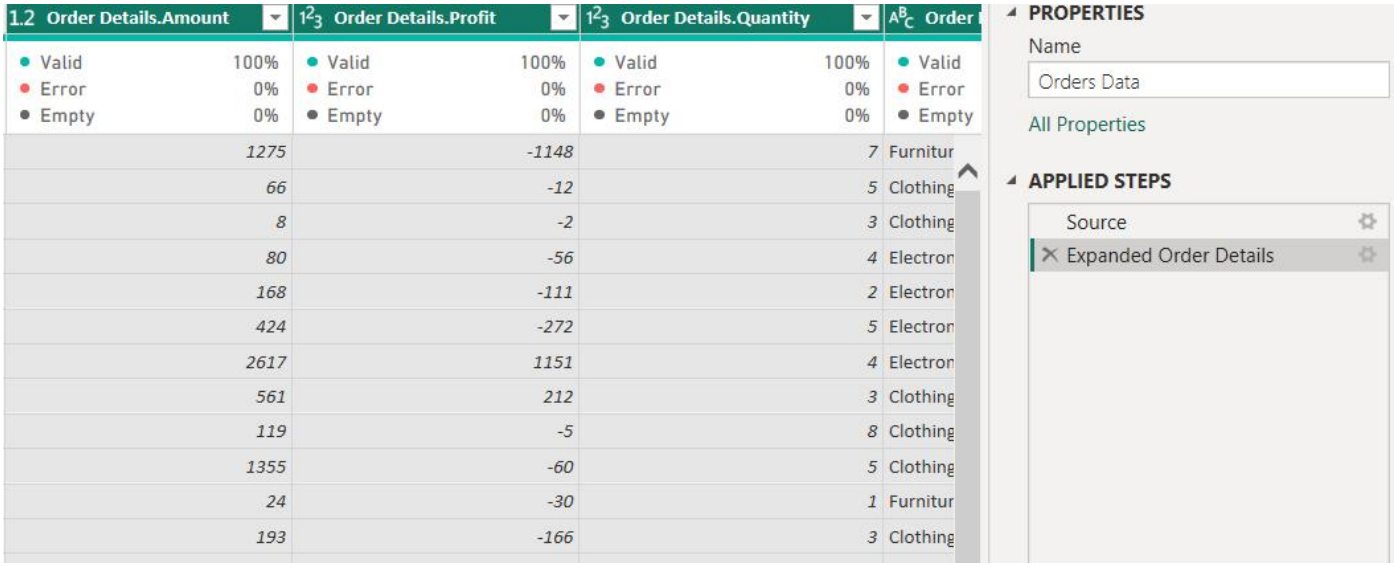
■ Profit > 0 → Profit



6. Merging Data (Joins)

○ Merge the List of Orders and Order Details tables using Order ID.

○ Name the merged output table as Orders Data.



1.2 Order Details.Amount	123 Order Details.Profit	123 Order Details.Quantity	ABC Order
Valid 100%	Valid 100%	Valid 100%	Valid
Error 0%	Error 0%	Error 0%	Error
Empty 0%	Empty 0%	Empty 0%	Empty
1275	-1148	7	Furnitur
66	-12	5	Clothing
8	-2	3	Clothing
80	-56	4	Electron
168	-111	2	Electron
424	-272	5	Electron
2617	1151	4	Electron
561	212	3	Clothing
119	-5	8	Clothing
1355	-60	5	Clothing
24	-30	1	Furnitur
193	-166	3	Clothing

PROPERTIES

Name

Orders Data

All Properties

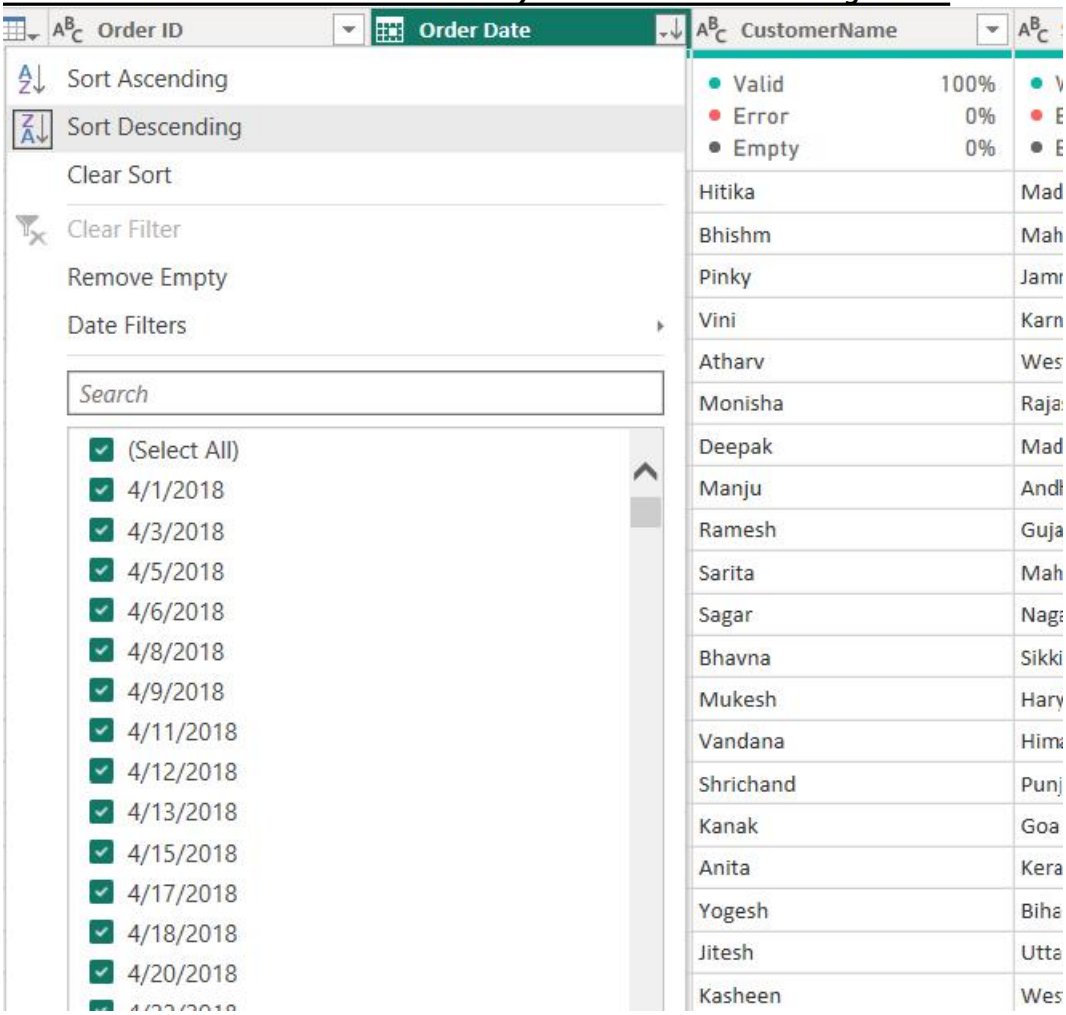
APPLIED STEPS

Source

Expanded Order Details

8. Sorting and Filtering Data

● **Sort records in the Orders Data table by Order Date in descending order.**



● **Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).**

Order ID	Order Date	CustomerName	State	City	Location	Amount
B-26081	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-26018	2/14/2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-26008	2/9/2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25860	11/15/2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25788	9/21/2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25716	7/11/2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25698	6/23/2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	

9. Grouping and Aggregating Data

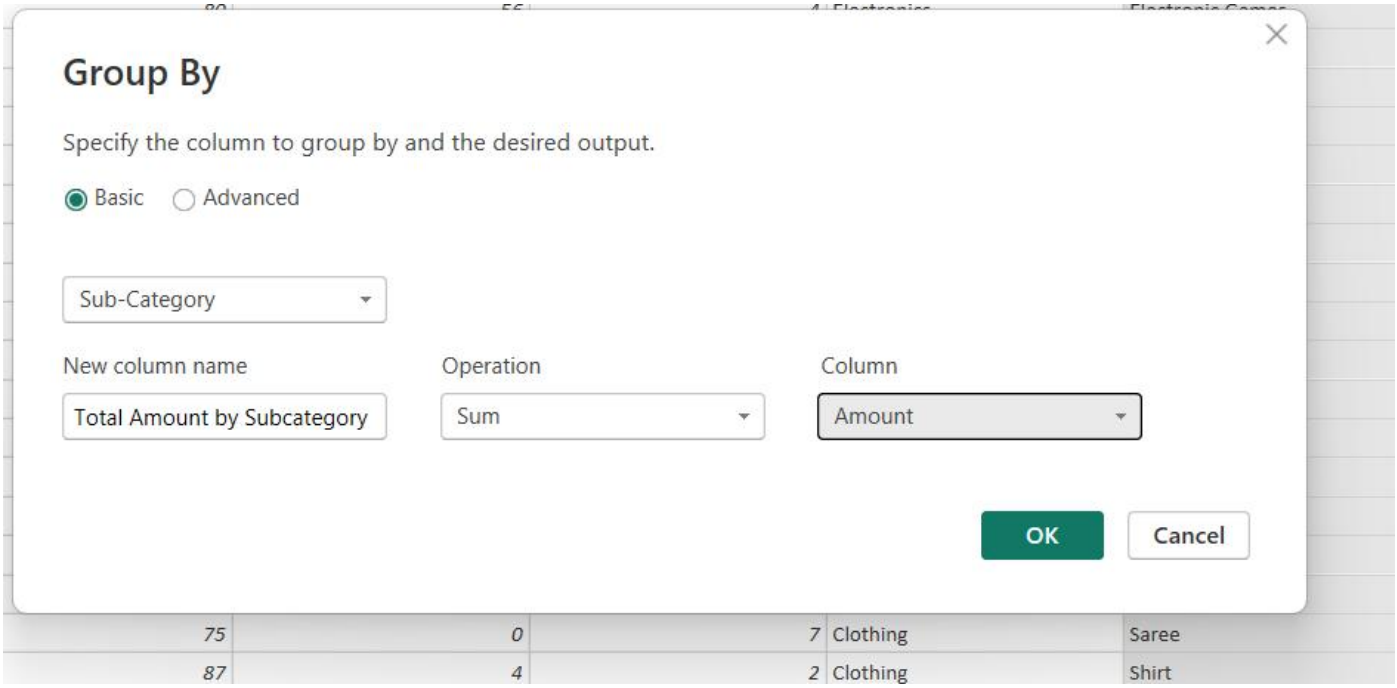
● **Duplicate the Order Details table and perform the following: ○ Count of Order ID**

	Order ID	Count of Order ID
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
	B-25601	4
	B-25602	5
	B-25603	8
	B-25604	2
	B-25605	1
	B-25606	1
	B-25607	1
	B-25608	4
	B-25609	2
0	B-25610	6
1	B-25611	1
2	B-25612	1
3	B-25613	1
4	B-25614	2
5	B-25615	1
6	B-25616	4
7	B-25617	1

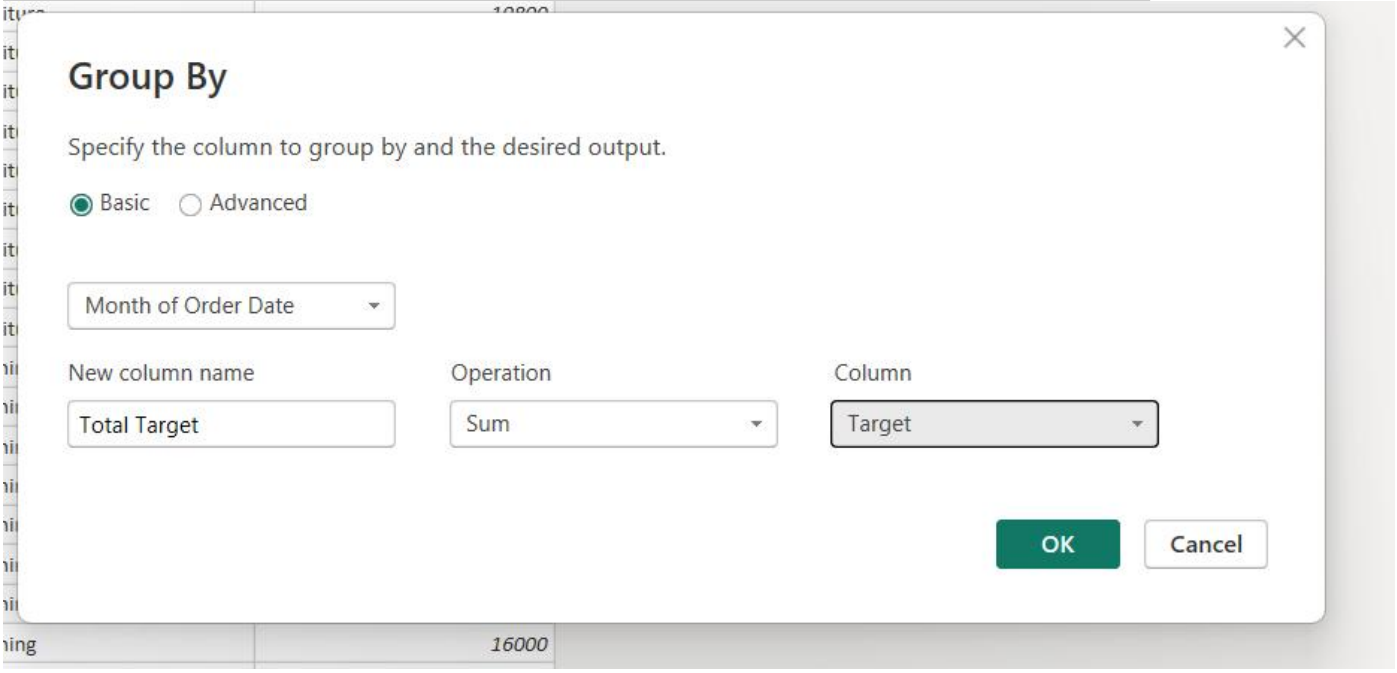
○ ***Average Profit by Category***

	Category	1.2 Avergae Profit of category
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
1	Furniture	9.456790123
2	Clothing	11.76290832
3	Electronics	34.07142857

○ ***Total Amount by Sub-Category***



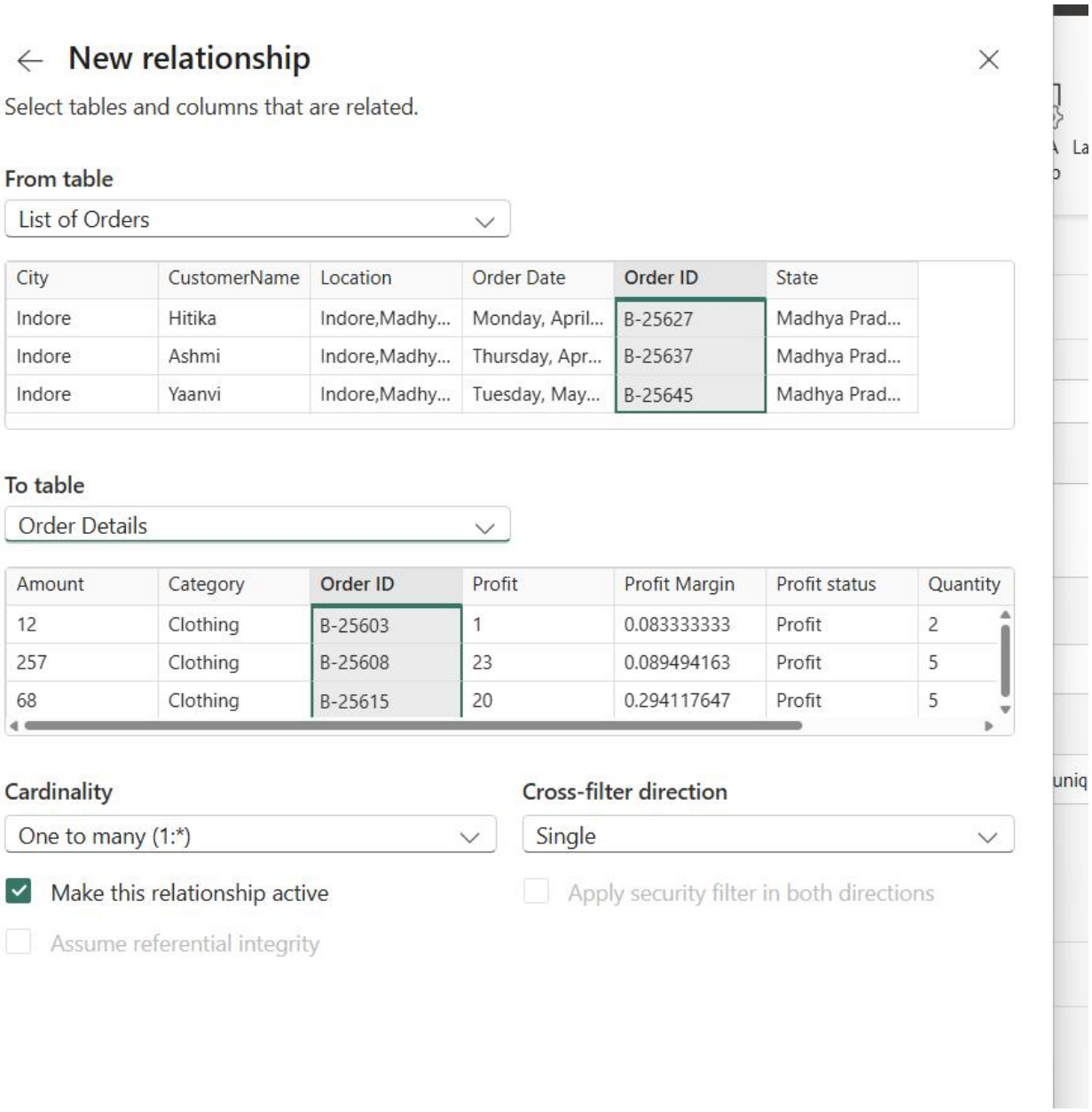
● Duplicate the Sales Target table and aggregate the total Target by Month of order date.



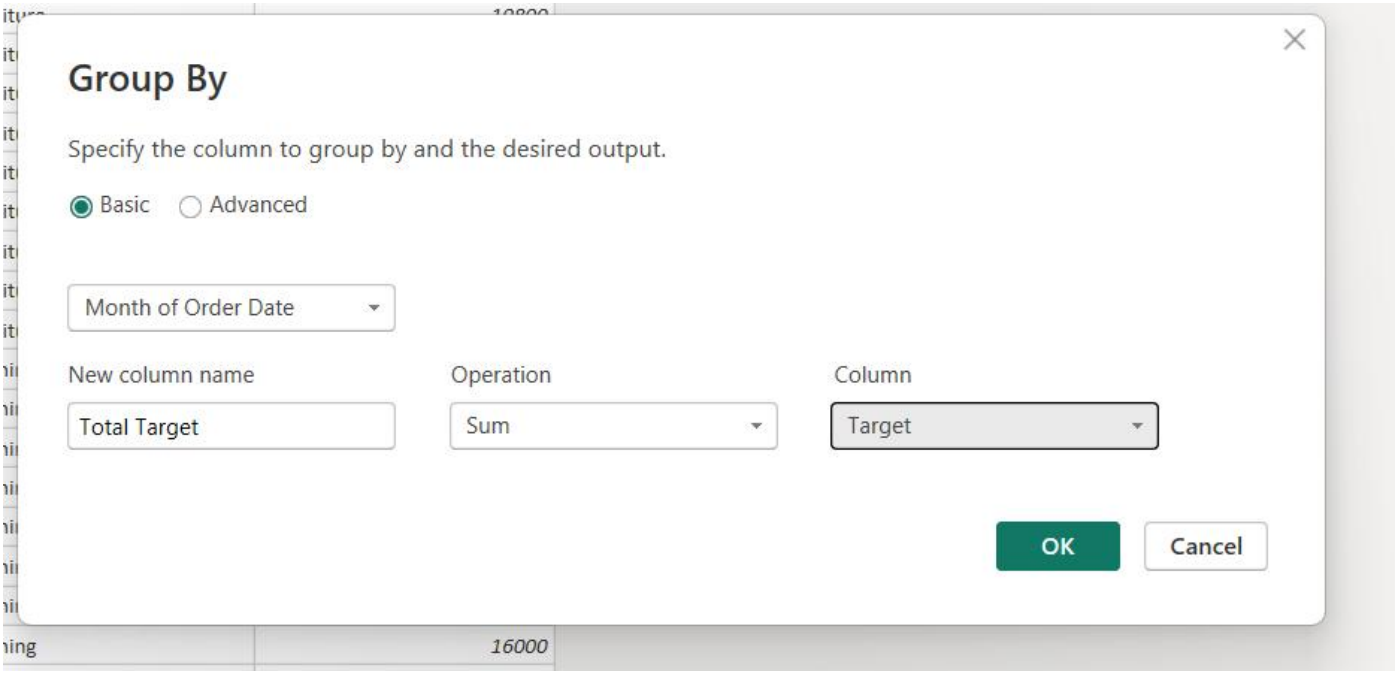
10. Data Modeling:

● Establish a relationship between:

○ List of Orders and Order Details using Order ID



- ***Establish a relationship between:***
  - ***Order Details and Sales Target using Category***



2. Use Manage Relationships to ensure relationships are active and correctly defined.

